

APTIVA

Intelligence that adapts to you.

SRS Document — IEEE 830-1998

Software Requirements Specification

Aptiva Adaptive E-Learning Platform

Project	Design and Implementation of an Adaptive E-Learning Platform based on Quality of Experience for Inclusive Online Education in Low-Resource Settings
App Name	Aptiva
Tagline	"Intelligence that adapts to you."
Course	CEF 440: Internet Programming and Mobile Programming
Institution	University of Buea — Faculty of Engineering and Technology
Department	Department of Computer Engineering
Instructor	Dr. Nkemeni Valery
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AFUH CHIKE CHEWAKONDI	FE23A005
PIGHIEBWOH NOKA GILDAS	FE23A135
TAKANG BRANDON EYONG	FE23A149

TAKU YVETTE WUTOH	FE23A152
TAMANTENG ABERLINE NGUEMUTANG	FE23A154

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for **Aptiva**, an intelligent adaptive mobile e-learning platform engineered for students in low-resource connectivity environments. This document conforms to the **IEEE 830-1998** Recommended Practice for Software Requirements Specifications.

This document is the authoritative backing specification for all design decisions across Tasks 1–7 of CEF 440. Every diagram in Task 4, every screen in Task 5, every table in Task 6, and every feature demonstrated in Task 7 can be traced directly to a requirement stated in this document.

WHY	Aptiva exists because standard e-learning platforms assume stable broadband. In Cameroon and Sub-Saharan Africa, students routinely experience bandwidths below 200 kbps, making those platforms unusable. Aptiva was engineered to learn around the network — not require the network.
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1.2 Project Scope

Aptiva is a Flutter-based cross-platform mobile application (Android/iOS) that dynamically adapts content delivery format based on real-time network quality measurement, optimising the learner's Quality of Experience (QoE) regardless of connection strength.

In-scope functionalities:

- Role-based user authentication (Student, Instructor, Administrator) with JWT and OAuth2
- Course content management — upload, transcode, categorise, and deliver in adaptive formats

- Real-time network quality monitoring — bandwidth, latency, packet loss, jitter, RSSI
- Adaptive Delivery Engine — automatic switching: HD video → SD → audio-only → offline cache
- Quality of Experience (QoE) assessment, feedback collection, and feedback-loop recalibration
- AI-powered personalised content recommendation engine (on-device ML inference)
- Offline content download, SQLite caching, and background synchronisation
- Interactive quiz delivery — offline attempt, auto-save every 30s, and background sync
- Instructor analytics dashboard with student progress monitoring and push/SMS alerts
- Live lecture broadcasting via WebRTC (Agora SDK) with adaptive audio fallback

Out-of-scope: Payment processing, third-party LMS integration, hardware procurement, content licensing.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Aptiva	The adaptive e-learning mobile application — the product defined in this SRS
QoE	Quality of Experience — subjective measure of the learner's perceived platform performance
QoS	Quality of Service — objective network metrics: bandwidth, latency, packet loss, jitter
ABR	Adaptive Bitrate Streaming — automatically adjusting video quality based on available bandwidth

HLS	HTTP Live Streaming — streaming protocol for adaptive video delivery using .m3u8 manifest files
JWT	JSON Web Token — signed authentication token containing identity and role claims
SQLite	Embedded relational database for local offline storage on the student's device
MOS	Mean Opinion Score — standardised 1-5 scale for subjective user experience rating
RSSI	Received Signal Strength Indicator — measures mobile network signal in dBm
FCM	Firebase Cloud Messaging — push notification delivery service
RBAC	Role-Based Access Control — restricts system access by assigned user roles
CDN	Content Delivery Network — distributed infrastructure for fast media delivery
HLS	HTTP Live Streaming — adaptive streaming protocol generating .m3u8 and .ts files
WebRTC	Web Real-Time Communication — peer-to-peer protocol for live audio/video sessions
QoS Classifier	Sub-process 4.1 of the Adaptive Delivery Engine — classifies network into quality bands
Hysteresis Filter	Algorithm preventing rapid quality band oscillation by requiring sustained state changes
SRS	Software Requirements Specification — this document

1.4 References

1. IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications
2. Mobile Programming Projects 2025–2026, CEF 440, University of Buea — Project 1
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4. Sommerville, I. (2015). Software Engineering, 10th Ed. Pearson
5. Booch, G., Rumbaugh, J., Jacobson, I. (2005). The UML User Guide. Addison-Wesley
6. Flutter Documentation: <https://docs.flutter.dev>
7. Firebase Documentation: <https://firebase.google.com/docs>
8. Agora WebRTC SDK: <https://docs.agora.io>

2. Overall System Description

2.1 Product Perspective

Aptiva is an independent mobile application that communicates with a multi-tier backend infrastructure. It does not replace existing LMS systems but rather provides an adaptive delivery layer on top of standard e-learning content. The system architecture consists of four tiers: Client Devices (Tier 1), Application Servers (Tier 2), Data Tier (Tier 3), and Cloud/External Services (Tier 4) — all modelled in detail in the Task 4 Deployment Diagram.

2.2 User Classes and Characteristics

User Class	Description	Technical Level	Primary Goals
Student	Primary learner — may own low-cost Android device; intermittent connectivity; daily usage	Low–Medium	Access course content, attempt quizzes, download for offline, track progress
Instructor	Content creator — uploads lectures and quizzes;	Medium	Upload materials, create quizzes, monitor student

	monitors student analytics		dashboards, send reminders
Administrator	System manager — manages accounts and platform policies	High	Account management, compliance, role assignment
Developer	System maintainer — runs tests and CI/CD deployments	Expert	Backend access, testing, monitoring, deployment

2.3 Operating Environment

2.3.1 Client Tier

- Platform: Android 8.0+ (API Level 26) and iOS 13.0+
- Framework: Flutter 3.x compiled with Dart 3.x
- Minimum RAM: 2 GB (4 GB recommended for HD playback)
- Minimum available storage: 4 GB (for offline content cache)
- Network: 2G / 3G / 4G / 5G / WiFi — Airtiva must function at 90 kbps minimum in audio-only mode

2.3.2 Server Tier

- Application Server: Node.js 20 LTS / Django REST Framework 4.x — Docker containerised
- Media Server: NGINX 1.25+ with FFmpeg 6.x — Docker containerised
- Database: PostgreSQL 15+ on port 5432; Redis 7+ on port 6379
- Cloud: AWS EC2 t3.medium or GCP Compute Engine — behind auto-scaling load balancer
- Object Storage: AWS S3 / GCP Cloud Storage for all video, audio, and PDF assets
- Authentication: Firebase Auth + OAuth 2.0 (Google Sign-In)
- Notifications: Firebase Cloud Messaging (FCM)

2.4 Design and Implementation Constraints

- All video streaming must implement HLS (HTTP Live Streaming) with .m3u8 manifests and .ts segments
- Offline data persistence must use SQLite on device — no third-party local database
- JWT access tokens expire after 24 hours; refresh tokens expire after 30 days
- All API communication must use HTTPS / TLS 1.3 — unencrypted HTTP is strictly prohibited
- Passwords must be hashed with bcrypt at minimum cost factor 12 before storage
- Maximum single video file upload size: 2 GB — chunked using tus.io resumable protocol
- ML recommendation model must support on-device inference — no cloud dependency required
- All UI must comply with WCAG 2.1 AA accessibility standard (4.5:1 contrast minimum)
- The app must operate in audio-only mode at minimum 90 kbps — the floor of 2G connectivity

2.5 Assumptions and Dependencies

- Students have access to at least periodic internet for initial content download and sync
- Firebase project is configured with valid API keys before app deployment
- Agora.io account and App ID are provisioned for WebRTC live lecture functionality
- FFmpeg is pre-installed on the media server Docker container image
- YouTube Data API v3 quota is sufficient for the concurrent student count

3. Functional Requirements

3.1 Authentication and Identity Management

FR-AUTH-01: Student and Instructor Registration

Description: Aptiva shall allow new users to self-register by providing full name, email address, password (minimum 8 characters, at least 1 uppercase, 1 number, 1 special character), and role selection (Student or Instructor).

Processing: Validate all fields server-side. Check email uniqueness against D1 (User Database). Hash password with bcrypt (cost 12). Create user record. Issue JWT access token and refresh token. Return to role-appropriate dashboard.

Error handling: 409 Conflict for duplicate email. 422 for validation failures. 400 for malformed input.

FR-AUTH-02: Secure Login with JWT

Aptiva shall authenticate users via email/password. On success, issue a JWT access token (24h) and refresh token (30d), stored on device using Flutter Secure Storage (iOS Keychain / Android Keystore). Biometric authentication (fingerprint/face ID) shall be supported as a secondary login method after initial password login.

FR-AUTH-03: Role-Based Access Control (RBAC)

All protected API endpoints shall validate JWT tokens and role claims before processing. Students access only learning features. Instructors access content management and analytics. Administrators access account management. Unauthorised access attempts return 403 Forbidden.

FR-AUTH-04: Offline Authentication Fallback

When the network is unavailable during login, Aptiva shall offer SMS OTP-based account recovery as an offline-capable fallback. Cached session tokens shall allow authenticated app access for up to 30 days without re-login.

3.2 Course Content Management

FR-COURSE-01: Resumable Instructor Upload

Instructors shall upload lecture videos (MP4, MOV, AVI — max 2 GB), PDF notes, and quiz JSON files using the tus.io resumable upload protocol. Upload progress shall display in real time as a percentage. Interrupted uploads shall resume automatically from the last successful chunk on reconnection.

FR-COURSE-02: Automated Multi-Format Transcoding

Upon upload completion, Aptiva's media server shall automatically transcode video files using FFmpeg into the following four adaptive formats:

Format	Resolution	Bitrate	File Type	Target Band
HD	1080p	5 Mbps	.m3u8 / .ts	Excellent
SD	480p	1 Mbps	.m3u8 / .ts	Good
Low	240p	300 kbps	.m3u8 / .ts	Poor (video)
Audio-only	N/A	48 kbps	.aac	Poor / Offline

FR-COURSE-03: Course Browsing, Enrolment, and Progress Tracking

Students shall browse all available courses, view course detail pages (title, description, instructor, lecture list, quiz count, format availability badges), and enrol or unenrol. Enrolment and progress data shall be persisted to D2 (Course Database) and synchronised on reconnection when modified offline.

FR-COURSE-04: Multi-Format Content Accessibility

Every lecture shall be accessible in at minimum two formats. The UI shall display format availability badges (HD / SD / Audio / Offline) per lecture. Content shall never be entirely unavailable — at minimum, audio-only or offline cached versions shall always be served.

3.3 Network Quality Monitoring

FR-NET-01: Continuous Real-Time Measurement

The Aptiva Network Monitor Agent running on the student device shall measure the following parameters every 2 seconds during active content consumption: bandwidth (Mbps), round-trip latency (ms), packet loss (%), jitter (ms), and signal strength (RSSI, dBm). All measurements are logged with timestamps to D3 (Network Log).

FR-NET-02: QoS Band Classification

The QoS Classifier (Sub-process 4.1 of the Adaptive Delivery Engine) shall compute a weighted composite network quality score and classify it into one of four bands:

Band	Score	Bandwidth	Latency	Packet Loss	Content Delivered
Excellent	80–100	> 5 Mbps	< 50ms	< 1%	HD 1080p video
Good	55–79	1–5 Mbps	50–150ms	1–3%	SD 480p video
Poor	25–54	200kbps–1Mbps	150–400ms	3–10%	Audio-only stream
Offline	0–24	< 200 kbps or none	N/A	N/A	D6 cache served

FR-NET-03: Hysteresis-Controlled Switching

To prevent disruptive rapid quality oscillation, the hysteresis filter shall require: **10 consecutive seconds** in a higher band to trigger an upgrade; **3 consecutive seconds** in a lower band to trigger a downgrade. Downgrades are faster because stalling causes immediate disruption; upgrades are slower to confirm the improvement is stable.

3.4 Adaptive Content Delivery Engine

FR-ADAPT-01: Seamless Dynamic Format Switching

The Adaptive Delivery Engine (Process 4.0) shall continuously evaluate the current quality band and switch delivery format without requiring any student action. Format transitions shall complete within 2 seconds with no playback interruption exceeding 2 seconds (NFR-PERF-02).

FR-ADAPT-02: Audio-Only Mode with Student Notification

When the quality band drops to Poor, Aptiva shall automatically switch to audio-only streaming. A **rose-coloured alert banner** shall appear: 'Low bandwidth detected — switched to audio-only mode.' The audio waveform visualisation shall confirm active audio playback. When bandwidth recovers to Good or Excellent for 10 sustained seconds, a 'Bandwidth restored — tap to resume video' option shall appear with a Resume button.

FR-ADAPT-03: Offline Mode with Full Continuity

When connectivity is entirely lost (Offline band), Aptiva shall immediately switch to serving content from D6 (Video/Media Cache — local SQLite). Student interactions (quiz answers,

watch progress, QoE feedback) shall be queued with timestamps and synchronised to the backend when connectivity is restored via the background sync service.

3.5 Quiz Management

FR-QUIZ-01: Offline-First Quiz Delivery

All quiz question sets shall be pre-downloaded to the device in JSON format when the student first enrolls in a course. Questions shall be served from local SQLite storage — internet is not required to attempt a quiz. Student progress auto-saves locally every 30 seconds.

FR-QUIZ-02: Background Sync on Reconnection

When internet connectivity is restored, the Aptiva sync service shall push all pending quiz answers to the backend Quiz API with original timestamps preserved. Scores and instant feedback JSON shall be returned and displayed immediately. If a student completes a quiz offline and another student has submitted the same quiz online in the interim, conflict resolution shall favour the offline timestamp.

FR-QUIZ-03: Instructor Quiz Builder

Instructors shall create quizzes with three question types: Multiple Choice (MCQ), True/False (T/F), and Short Answer. Each question shall have defined answer keys and optional explanation text. Published quizzes shall be pushed to all enrolled student devices as JSON for offline caching via FCM.

3.6 QoE Assessment and Feedback Loop

FR-QOE-01: Subjective Session Feedback

After each content session, Aptiva shall prompt students with a QoE feedback screen collecting: star rating (1–5), slider scores for Video Quality, Loading Speed, and Content Clarity (each 0–10), and an optional text comment. Session metadata tags (e.g. 'Offline mode used', 'N quality switches') shall be auto-generated and appended.

FR-QOE-02: Objective QoE Event Logging

Aptiva shall automatically log the following objective QoE events per session to D4 (QoE Metrics Store): number of quality band switches, buffer underrun events, total audio-only duration (seconds), time-to-first-frame, session completion percentage, and network band distribution (% time in each band).

FR-QOE-03: Adaptive Threshold Recalibration

Collected QoE scores shall feed back into the Adaptive Delivery Engine via Process 6.0 (QoE Assessment). If a statistically significant number of students report poor experience

(MOS < 2.5) at the 'Good' band, the QoS Classifier thresholds for that student's device class shall be automatically recalibrated to treat lower bandwidths as 'Poor'.

3.7 AI-Powered Content Recommendation

FR-REC-01: Personalised Recommendation Engine

The Aptiva recommendation engine shall generate personalised content suggestions based on: enrolment history, quiz performance scores, lecture watch-time percentage, QoE satisfaction scores, and content similarity vectors. Recommendations shall appear on the student dashboard as 'Recommended by AI' with a violet AI badge.

FR-REC-02: On-Device Inference Capability

The recommendation model (stored in D5 as a compressed .pkl or TensorFlow Lite .tflite file) shall support fully offline inference on the student device. Recommendations shall remain available even without internet access, using the last-synced model weights.

3.8 Instructor Analytics Dashboard

FR-ANALYTICS-01: Comprehensive Progress Metrics

The Aptiva instructor dashboard shall display: per-student completion rates, average quiz scores, lecture watch-time per student, QoE satisfaction averages, network quality distribution, and a highlighted list of students with completion rates below 30% (displayed with rose alert badge).

FR-ANALYTICS-02: Push Notification and SMS Reminders

From the dashboard, instructors shall send targeted FCM push notifications or SMS reminders to individual students or cohorts. Messages shall be composable inline. Delivery receipts shall be logged and visible to the instructor.

3.9 Live Lecture (WebRTC)

FR-LIVE-01: Agora WebRTC Session

Instructors shall initiate live class sessions via the Agora WebRTC SDK. Students join via shared session link received through FCM notification. The session shall support: screen sharing, whiteboard overlay, simultaneous multi-student viewing, session recording, and post-session replay.

FR-LIVE-02: Adaptive Audio Fallback for Live Sessions

During live sessions, if a student's bandwidth drops below 512 kbps, their incoming stream shall automatically downgrade to audio-only. The instructor view shall show each student's

real-time connection quality as a coloured indicator dot. Students in audio-only mode shall receive a banner notification.

4. Non-Functional Requirements

4.1 Performance

ID	Requirement	Target
NFR-PERF-01	Time-to-first-frame (video load)	< 3s on Good band; < 6s on Poor band
NFR-PERF-02	Quality switch latency	< 2 seconds — no visible playback stall
NFR-PERF-03	REST API response time	< 500ms at P95 under 500 concurrent users
NFR-PERF-04	Offline SQLite query	< 100ms for any local content query
NFR-PERF-05	Quiz auto-save	< 50ms per local write
NFR-PERF-06	Background sync after reconnect	< 30 seconds for all pending quiz answers
NFR-PERF-07	App cold-start to dashboard	< 4 seconds on minimum-spec device
NFR-PERF-08	FCM push notification delivery	< 10 seconds to device
NFR-PERF-09	Recommendation inference (on-device)	< 200ms per recommendation batch

4.2 Security

- NFR-SEC-01: All passwords hashed with bcrypt cost factor 12 minimum — never stored in plaintext
- NFR-SEC-02: All client-server communication over HTTPS/TLS 1.3 — no HTTP fallback
- NFR-SEC-03: JWT tokens stored in Flutter Secure Storage (iOS Keychain / Android Keystore Encrypted SharedPreferences)
- NFR-SEC-04: API rate limiting — 100 requests/minute per IP; 1000/minute per authenticated user
- NFR-SEC-05: All database queries use parameterised statements — SQL injection prevention
- NFR-SEC-06: CORS policy restricted to registered Aptiva application origins only
- NFR-SEC-07: Video/PDF content URLs signed with time-limited access tokens (4-hour expiry)
- NFR-SEC-08: Offline SQLite cache encrypted with AES-256 using device-bound key
- NFR-SEC-09: Sensitive PII (email, password hash) excluded from all analytics logs

4.3 Reliability and Availability

- NFR-REL-01: Backend service uptime target — 99.5% monthly availability
- NFR-REL-02: Offline mode must remain fully functional for minimum 72 hours without any internet connection
- NFR-REL-03: Failed sync retries use exponential backoff — 1s, 2s, 4s, 8s, 16s, max 64s
- NFR-REL-04: All database writes use transactions with automatic rollback on failure
- NFR-REL-05: PostgreSQL database backed up daily to S3 with 30-day retention
- NFR-REL-06: Redis cache miss must gracefully fall back to PostgreSQL — no request failure

4.4 Usability

- NFR-USE-01: A new student must register, find a course, and start watching within 5 minutes without training
- NFR-USE-02: Minimum readable text size: 12sp — no smaller text anywhere in the app
- NFR-USE-03: Network quality indicator must always be visible during content consumption — persistent badge
- NFR-USE-04: WCAG 2.1 AA compliance — minimum 4.5:1 contrast ratio for all text
- NFR-USE-05: All primary student actions must be reachable with right thumb on a 6-inch screen
- NFR-USE-06: Offline content must be clearly distinguished from online content via UI badges
- NFR-USE-07: Every loading state must use shimmer skeleton — never blank white screen

4.5 Scalability

- NFR-SCALE-01: Auto-scaling group must handle 10x normal traffic within 60 seconds of spike detection
- NFR-SCALE-02: PostgreSQL connection pool — maximum 100 concurrent connections
- NFR-SCALE-03: Redis cluster read latency < 5ms under 10,000 concurrent sessions
- NFR-SCALE-04: S3 object storage must support concurrent streaming to 5,000+ students without throttling

4.6 Maintainability

- NFR-MAINT-01: Flutter code follows Clean Architecture — Presentation / Domain / Data layer separation
- NFR-MAINT-02: Unit test coverage minimum 70% for all business logic (domain layer)
- NFR-MAINT-03: All REST API endpoints documented via OpenAPI 3.0 specification
- NFR-MAINT-04: Docker containers support zero-downtime rolling updates

- NFR-MAINT-05: All environment variables managed via .env files — no hardcoded secrets in source

5. External Interface Requirements

5.1 Software Interfaces

External System	Interface	Purpose	Protocol
Firebase Auth	REST / FlutterFire SDK	User authentication and token validation	HTTPS / OAuth 2.0
Firebase Cloud Messaging	REST / FlutterFire SDK	Push notifications to students and instructors	HTTPS / FCM v1 API
YouTube Data API v3	REST	Video embedding and tutorial playlist queries	HTTPS / REST
AWS S3 / GCP Storage	REST SDK (Dart)	Store and retrieve video, audio, and PDF assets	HTTPS / S3 API
Agora WebRTC SDK	Native SDK (agora_rtc_engine)	Live lecture peer-to-peer audio/video sessions	WebRTC / DTLS-SRTP
Redis	TCP connection pool	Session store, metadata cache, task queue	TCP :6379
PostgreSQL	TCP connection pool	Primary relational database for all persistent data	TCP :5432
FFmpeg	Docker CLI / HTTP trigger	Video transcoding after instructor upload	Internal HTTP

5.2 Communication Interfaces

- Client-to-server: HTTPS/TLS 1.3 on port 443 for all REST API calls
- Video streaming: HLS over HTTPS — .m3u8 manifest + .ts segment delivery
- Live sessions: WebRTC with DTLS-SRTP encryption via Agora SDK
- Database: TLS-encrypted TCP — PostgreSQL :5432, Redis :6379
- Push notifications: FCM HTTP v1 API over HTTPS with OAuth 2.0 service account
- Background sync: Platform WorkManager (Android) / Background Tasks (iOS) via workmanager package

6. Requirements Traceability Matrix

Requirement ID	Feature	Covered In	Test Method
FR-AUTH-01 – 04	User Authentication	Task 4: DFD 1.0, Seq. Auth Flow; Task 5: Login/Register screens; Task 6: users_table	Unit test + Integration test + Manual UI test
FR-COURSE-01 – 04	Course Content Management	Task 4: DFD 2.0, Seq. Upload Flow; Task 5: Course screens; Task 6: courses_table	Integration + Manual upload test
FR-NET-01 – 03	Network Monitoring	Task 4: DFD 3.0, Sub-process 4.1; Task 5: Network badge widget	Unit test + Network throttle simulation
FR-ADAPT-01 – 03	Adaptive Delivery Engine	Task 4: DFD 4.0 (core), Seq. Adaptive Streaming; Task 5: Video/Audio-only screens	Integration + Bandwidth throttle test

FR-QUIZ-01 – 03	Quiz Management	Task 4: Use Case, Seq. Quiz Flow; Task 5: Quiz screen; Task 6: quiz tables	Offline test + Background sync test
FR-QOE-01 – 03	QoE Assessment	Task 4: DFD 6.0, Seq. Feedback Flow; Task 5: Feedback screen	User acceptance test + Automated logging
FR-REC-01 – 02	Recommendation Engine	Task 4: DFD 7.0; Task 5: Dashboard AI section	ML model evaluation + On-device inference test
FR-ANALYTICS-01 – 02	Instructor Analytics	Task 4: Use Case; Task 5: Instructor Dashboard; Task 6: analytics views	Manual test + FCM delivery verification
FR-LIVE-01 – 02	Live Lecture (WebRTC)	Task 5: Broadcaster screen; Task 4: Seq. Live Class	Agora WebRTC integration test + Adaptive test
NFR-PERF-01 – 09	All Performance NFRs	Task 5: All screens; Task 6: Query optimisation; Task 7: Performance demo	Load test (k6) + Device timing measurement
NFR-SEC-01 – 09	All Security NFRs	Task 4: Deployment Diagram (HTTPS/TLS); Task 6: bcrypt/JWT implementation	OWASP ZAP scan + Penetration test checklist

7. Appendices

Appendix A — Task Timeline

Task	Title	Marks	Due Date
Task 1	Mobile App Development Process	10 marks	08 April 2023
Task 2	Requirement Gathering	10 marks	15 April 2023
Task 3	Requirement Analysis	10 marks	15 April 2023
Task 4	System Modelling and Design	10 marks	13 May 2023
Task 5	UI Design and Implementation	10 marks	27 May 2023
Task 6	Database Design and Implementation	10 marks	03 June 2023
Task 7	Final Project Presentation	40 marks	18 June 2023

Appendix B — Network Quality Thresholds

Parameter	Excellent	Good	Poor	Offline
Bandwidth	> 5 Mbps	1–5 Mbps	200kbps–1Mbps	< 200 kbps
Latency (RTT)	< 50ms	50–150ms	150–400ms	Timeout
Packet Loss	< 1%	1–3%	3–10%	> 10%
Jitter	< 10ms	10–30ms	> 30ms	N/A
RSSI	> -70 dBm	-70 to -85 dBm	-85 to -100 dBm	< -100 dBm

Content Delivered	HD 1080p	SD 480p	Audio-only .aac	D6 SQLite cache
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